

Samuel Bondoc

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EXPERIENCE

Embedded Software Engineer

May 2023 - Present

Arkham Technology

Irvine, CA

- Led a team of 6 engineers in a 2-week sprint Agile environment in development of remote management capabilities on a satellite traffic encryption unit. [**Azure Devops**]
- Vastly improved project timeline by eliminating the need to rewrite 20,000 lines of code by designing a security boundary-compliant cross-processor cryptographic service. [**C** | **C++** | **OpenSSL** | **AMD Versal SoC** | **Rust**]
- Directly expanded the company's product portfolio by designing, implementing, and testing its whole CNSA 2.0 software library from scratch. [**C** | **SHA-3 (Keccak)** | **ML-KEM (CRYSTALS Kyber)** | **ML-DSA (CRYSTALS Dilithium)**]
- Decreased hardware testing time by 80% by automating software deployment process. [**POSIX shell** | **Bash** | **Docker**]
- Increased team velocity by eliminating build time errors with a clean dependency management API. [**CMake**]
- Streamlined source code synchronization between air-gapped networks by automating configuration management artifact generation. [**POSIX shell** | **Bash** | **Git** | **rsync** | **tar**]
- Decreased latency of remote command and control by 95% through implementation of session persistence and concurrent connections. [**HTTPS** | **RESTCONF** | **NETCONF**]
- Adapted end cryptographic unit runtime software to customer Linux platform by porting from Ubuntu x86-64 to Buildroot AArch64. [**Linux** | **GNU Core Utilities** | **GNU C Library** | **BusyBox** | **musl**]

Software Quality Assurance Engineer

June 2021 - May 2023

Toshiba America Business Solutions

Lake Forest, CA

- Collaborated with a team of 7 engineers in a 2 to 3-week sprint Agile environment.
- Managed suite of over 1000 test cases for cloud service printing system. [**Team Foundation Server (TFS)**]
- Developed automation for regression tests on virtual machines. [**C#** | **SQL** | **Selenium** | **Visual Studio** | **Hyper-V**]
- Built and deployed applications to 3 staging and 6 production environments. [**Azure** | **AWS** | **Linux** | **Git** | **Bash**]

SKILLS

Languages

- C
- C++
- POSIX shell
- Python
- CMake
- Bash
- Rust

Tools

- Git
- Docker
- GDB
- Valgrind
- VirtualBox
- Hyper-V

Environments

- POSIX
- Linux
- Windows
- Ubuntu
- Buildroot

Protocols

- HTTP
- TLS
- NETCONF
- RESTCONF
- TCP
- IP

Algorithms

- SHA-3 (Keccak)
- ML-KEM (Kyber)
- ML-DSA (Dilithium)

PROJECTS

IoT Irrigation System

- Designed irrigation prototype with DHT-11 sensor, HW-416 PIR sensor, and I2R LCD. [**Raspberry Pi** | **Python**]
- Optimized distribution efficiency protocol with CIMIS database retrieval over FTP.

Single-Cycle MIPS Processor HDL Model

- Implemented core, pipelining, and hazard protection functionalities. [**Verilog**]
- Tested and analyzed power consumption on simulated Kintex-7 FPGA. [**Xilinx Vivado**]

EDUCATION

University of California, Irvine

Bachelor of Science in Computer Engineering

Irvine, CA